Problem 12. TIER 3 — THERE IS NO PICTURE

Two resistors R_1 and R_2 are wired in parallel, so the total resistance R satisfies $\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2}$. At one instant $R_1 = 30\ \Omega$ and is rising at $2\ \Omega/\text{s}$, while $R_2 = 60\ \Omega$ and is rising at $3\ \Omega/\text{s}$. How fast is R changing?

(1) No picture

(2) want $\frac{dR}{dt}$, know $\frac{dR_1}{dt} = 2$, $\frac{dR_2}{dt} = 3$, instant $R_1 = 30$, $R_2 = 60$

(4) ~~(3)~~ $\frac{d}{dt}\left(\frac{1}{R}\right) = \frac{d}{dt}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

$\frac{d}{dt} R^{-1} = \frac{d}{dt} R_1^{-1} + \frac{d}{dt} R_2^{-1}$

$-1(R)^{-2}\frac{dR}{dt} = \ldots$

$-\frac{1}{R^2}\frac{dR}{dt} = -\frac{1}{R_1^2}\frac{dR_1}{dt} - \frac{1}{R_2^2}\frac{dR_2}{dt}$

(5) $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{2}{2}\cdot\frac{1}{30} + \frac{1}{60} = \frac{3}{60} = \frac{1}{20} \Rightarrow R = 20$

$\left(\frac{-1}{20^2}\right)\frac{dR}{dt} = \left(-\frac{1}{30^2}\,2 - \frac{1}{60^2}\cdot 3\right)(-400)$

$\frac{dR}{dt} = 400\left(\frac{2\cdot 4}{900\cdot 4} + \frac{3}{3600}\right) = 400\left(\frac{11}{3600}\right) = \frac{11}{900}\cdot 400$

$= \frac{11}{9}\ \frac{\Omega}{s} \simeq 1.22\ \Omega/s$